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Solid State Electronics – ELE717

by

ENGR. MRS RUBY CHINYERE BEREMEH ONYESOH

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15-Week Lecture Breakdown

Module 1: Material Physics & Semiconductor Growth (Weeks 1–3)

Week 1: Introduction to Semiconductor Materials

Elemental vs. compound semiconductors (Si, Ge, GaAs).

Crystal structures, Miller indices, and bonding mechanisms.

Week 2: Material Growth Techniques

oPurification of Silicon (metallurgical to electronic grade).

oCzochralski bulk crystal growth method.

oEpitaxial growth techniques (VPE, LPE, MBE).

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Week 3: Impurity Control & Doping

Diffusion and Ion Implantation mechanisms.

Controlled doping profiles and evaluation of impurity distribution.

Module 2: Energy Bands & Carrier Statistics (Weeks 4–6)

Week 4: Energy Bands & Carrier Concentrations Quantum mechanics foundations (Schrödinger equation overview, Kronig-Penney model).

Energy band gaps, allowed and forbidden states.

Intrinsic and extrinsic carrier concentrations.

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Week 5: Carrier Statistics

Fermi-Dirac distribution function.

Density of states, Fermi energy level calculation as a function of temperature and doping.

Week 6: Continuous Assessment Test 1 (CAT 1) & Review

Mid-module evaluation of crystal structures and carrier statistics.

Module 3: Carrier Transport & Injection Phenomena (Weeks 7–9)

Week 7: Drift and Diffusion Currents

Carrier mobility, scattering mechanisms, and conductivity.Diffusion of carriers, Fick's law, and Einstein relation.

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Week 8: High-Field Effects & Excess Carriers

Velocity saturation and hot carrier effects.

Generation and recombination mechanisms (Shockley-Read-Hall theory).

Carrier lifetime and steady-state excess carrier injection.

Week 9: Continuity Equation & Surface Phenomena

Derivation and application of the minority carrier continuity equation.

Semiconductor surface states and surface recombination velocity.

Module 4: Junctions, Interfaces, &.Characterisation (Weeks 10–12)

Week 10: The p-n Junction Diode Physics

Built-in potential, depletion width, and electric field profile.

Ideal diode equation (Shockley equation) derivation under forward and reverse bias.

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Week 11: Metal-Semiconductor Interfaces

Schottky barrier diode physics vs. Ohmic contact mechanisms.

Energy band diagrams under equilibrium and bias.

Week 12: Interface Characterisation & CAT 2

Methods of characterisation (Capacitance-Voltage.C-V profiling, Current-Voltage I-V analysis).

Continuous Assessment Test 2.

Module 5: Advanced Device Technology (Weeks 13–15)

Week 13: Bipolar Junction Transistors (BJTs)

Operational physics, minority carrier profiles in active regions.
Emitter efficiency, base transport factor, and current gains (α, β).

Week 14: Field-Effect Transistors (MOSFETs)
Ideal MOS capacitor physics: Accumulation, depletion, and inversion.
MOSFET threshold voltage, linear and saturation operating regions.

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Week 15: Revision & Final Exam Preparation

Overview of the entire course curriculum, problem-solving clinics, and past question reviews.

Laboratory & Practical Component (3 Hours/Week)

Students will complete at least 4 practical or simulation experiments:

1.Simulation of Carrier Transport: Modeling Fermi level shifts using MATLAB or Python.

2.I-V Characterisation: Measuring and plotting the forward/reverse characteristics of a p-n junction and a Schottky diode.

3.C-V Profiling: Determining doping concentration and built-in potential from capacitance measurements.

4.MOSFET Parameter Extraction: Measuring threshold voltage (\(V_{th}\)) and channel mobility.

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Introduction

- Semiconductor materials are the foundation of modern electronic and electrical systems. Almost every modern electronic device including computers, mobile phones, solar cells, sensors, communication systems, power converters, and control systems depends on semiconductor technology.
- A **semiconductor** is a material whose electrical conductivity lies between that of a good conductor and an insulator. More importantly, its electrical conductivity can be **controlled** by temperature, light, electric fields, magnetic fields, and the intentional addition of impurities known as **dopants**.

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The most important semiconductor materials used in electronics include:

- Silicon (Si)
- Germanium (Ge)
- Gallium Arsenide (GaAs)
- Gallium Nitride (GaN)
- Silicon Carbide (SiC)
- Indium Phosphide (InP)
- Gallium Phosphide (GaP)

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- **What Is a Semiconductor?** A **semiconductor** is a material that has electrical conductivity between that of a conductor and an insulator and whose conductivity can be significantly controlled by temperature, impurities, light, or applied electric fields. Examples include: Silicon (Si) Germanium (Ge) and Gallium arsenide (GaAs)
- The conductivity of a semiconductor is not fixed. This makes semiconductor materials extremely useful for electronic devices.
- For example, pure silicon at room temperature is relatively poorly conducting. However, when a small controlled quantity of an impurity such as phosphorus or boron is introduced, its electrical conductivity can increase dramatically.
- This process is called **doping**.

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Classification of Materials According to Electrical Conductivity

Materials can broadly be classified into:

- Conductors
- Semiconductors
- Insulators

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Conductors

- Conductors have a large number of free electrons available for electrical conduction.

Examples include: Copper, Aluminum, Silver, Gold, Iron

- Copper is widely used in electrical wiring because of its high electrical conductivity.
- **Characteristics**
- High electrical conductivity
- Low electrical resistivity
- Large number of free charge carriers
- Small or effectively overlapping energy gap

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Insulators : Insulators have very few free charge carriers under normal conditions. Examples include: Glass, Rubber, Plastic, Ceramic,Mica

- They strongly resist the flow of electric current.

Characteristics

- Very high resistivity
- Very low conductivity
- Large energy bandgap
- Electrons are strongly bound to atoms

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- **Semiconductors :** Semiconductors have conductivity between conductors and insulators. Examples: Silicon, Germanium, Gallium arsenide, Silicon carbide, Gallium nitride
- Their conductivity can be modified through: Temperature, Doping, Illumination, Electric field, Magnetic field, Mechanical stress
- This controllability is one of the most important characteristics of semiconductor materials.

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Comparison of Conductors, Semiconductors and Insulators

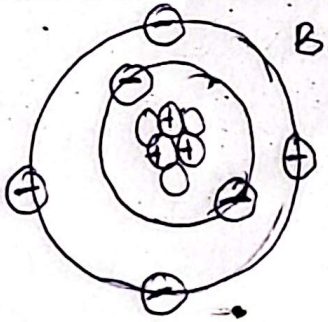
Property	Conductor	Semiconductor	Insulator
Conductivity	Very high	Moderate/controllable	Very low
Resistivity	Very low	Intermediate	Very high
Bandgap	Approximately zero	Small	Large
Free electrons	Many	Moderate/controllable	Very few
Temperature effect	Conductivity usually decreases	Conductivity usually increases	Small effect under normal conditions
Examples	Cu, Ag, Al	Si, Ge, GaAs	Glass, rubber, plastic
Main application	Wiring	Electronic devices	Electrical isolation

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Atomic Structure of Semiconductor Materials

To understand semiconductor materials, it is necessary to understand their atomic structure. An atom consists of:

- ❑ Nucleus
- ❑ Protons
- ❑ Neutrons
- ❑ Electrons



Bohr model method

⊖ electrons

⊕ proton

○ neutron

- ✓ The nucleus contains protons and neutrons, while electrons occupy different energy levels around the nucleus.
- ✓ The electrons in the outermost shell are particularly important in determining the electrical properties of a material.
- ✓ These are called **valence electrons**.

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- **Valence Electrons:** The electrons located in the outermost energy level of an atom are called **valence electrons**.

For example:

❑**Silicon:** Silicon has atomic number: $Z = 14$

- Its electron configuration is: $1s^2 2s^2 2p^6 3s^2 3p^2$
- Therefore, silicon has: 4 valence electrons.

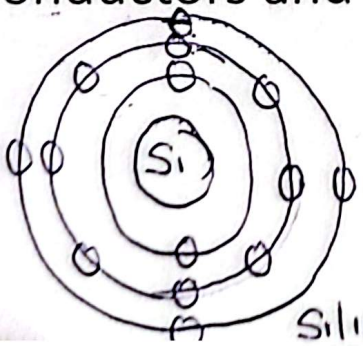
❑**Germanium** Germanium has atomic number: $Z = 32$
Its electron configuration is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$

- It also has four valence electrons.

❑**Gallium:** Gallium has three valence electrons.

❑**Arsenic:** Arsenic has five valence electrons.

This difference becomes important when forming compound semiconductors and when doping semiconductor materials.



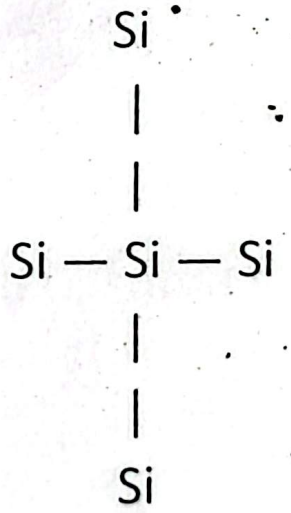
Atomic mass - 28.085
Electron Configuration - 2 8 4
Silicon

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Covalent Bonding in Silicon

Silicon atoms form a **covalent bonding structure**. Each silicon atom has four valence electrons. A silicon atom shares its four valence electrons with neighboring silicon atoms. Therefore, each silicon atom forms four covalent bonds.

Simplified representation:



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Cont.

- The shared electrons help hold the crystal structure together.
- At very low temperatures, most of these electrons remain involved in covalent bonds, resulting in very low electrical conductivity.
- When sufficient energy is supplied, some electrons can break away from their bonds and become mobile.

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Electron-Hole Pair Generation due to Thermal Excitation

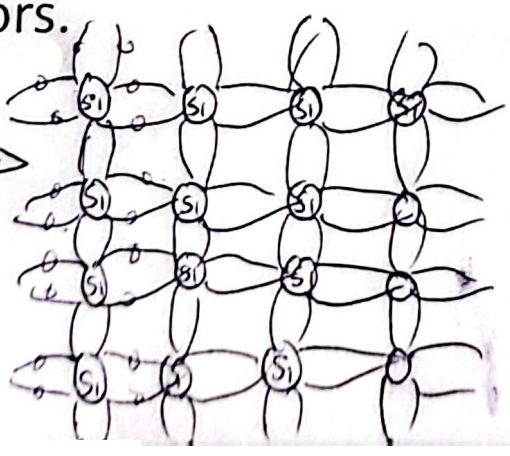
- When a valence electron receives sufficient energy, it can leave its covalent bond.
- The electron becomes a **free electron**.
- The absence left behind is called a **hole**.

Therefore:

$$\text{Energy } \textit{supplied} \rightarrow \textit{electron–hole pair}$$

- The electron has negative charge: $q = -1.6 \times 10^{-19} C$
- The hole behaves as if it has positive charge: $q = +1.6 \times 10^{-19} C$
- Both electrons and holes contribute to electrical conduction in semiconductors.

Electron-Hole pair →



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- **Intrinsic Semiconductor**

An **intrinsic semiconductor** is a pure semiconductor without intentionally introduced impurities. Examples:

- Pure silicon
- Pure germanium
- In an intrinsic semiconductor: $n = p = n_i$

where:

n= electron concentration

p= hole concentration

n_i =intrinsic carrier concentration

Therefore:

$$n = p$$

The number of electrons is approximately equal to the number of holes.

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Extrinsic Semiconductor

- An **extrinsic semiconductor** is a semiconductor whose electrical properties have been deliberately modified by adding controlled impurities. This process is known as: Doping

There are two major types:

- N-type semiconductor
- P-type semiconductor

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N-Type Semiconductor

An N-type semiconductor is produced by adding a **pentavalent impurity** to a semiconductor. Pentavalent atoms have five valence electrons. Examples include: Phosphorus (P), Arsenic (As), Antimony (Sb)

Suppose phosphorus is added to silicon.

- Phosphorus has five valence electrons, while silicon has four. Four electrons form covalent bonds with neighboring silicon atoms. The fifth electron is weakly bound and can easily become a free electron.

Thus, electrons become the majority carriers.

N-type material

Majority *carriers = electrons*	Minority *carriers = holes*

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- **P-Type Semiconductor**

A P-type semiconductor is produced by adding a **trivalent impurity**. Trivalent atoms have three valence electrons. Examples include: Boron (B), Aluminium (Al), Gallium (Ga), Indium (In)

When boron is introduced into silicon, only three covalent bonds can be completed. An electron is missing from the fourth bond. This missing electron is represented as a **hole**.

Therefore, holes become the majority carriers.

P-type material

Majority *carriers = holes*	Minority *carriers = electrons*

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Elemental Semiconductors

- An elemental semiconductor is made primarily from a single chemical element.

Important examples include:

Silicon (Si)--------- Atomic number: 14 ----------Four valence electrons

Widely used in integrated circuits

Used in power electronics

Used in solar cells

- **Germanium (Ge)**----------Atomic number: 32 ----------Four valence electrons

Higher carrier mobility than silicon

Lower bandgap than silicon

Used in specialized high-speed and infrared applications

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Compound Semiconductors

Compound semiconductors consist of two or more different elements. Examples include:

GaAs, GaN, InP, GaP, CdTe

- Compound semiconductors can provide properties that are difficult to achieve using elemental semiconductors. For example, GaAs has high electron mobility and is useful in high-frequency electronic systems, GaN has a wide bandgap and is important in: Power electronics, LEDs, RF systems, High-temperature electronics

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Silicon : Silicon is the most widely used semiconductor material in modern electronics.

- **Important properties**
- Atomic number: 14 ------Four valence electrons -------Diamond-cubic crystal structure
- Bandgap approximately: $E_g \approx 1.12\,eV$ ------- at room temperature.

Silicon is widely used for:

- Microprocessors
- Memory chips
- Diodes
- Transistors
- Solar cells
- Power devices
- Sensors
- Integrated circuits

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- **Germanium :** Germanium was one of the earliest semiconductor materials used extensively for transistor technology.

Important characteristics include:

- Four valence electrons ----------Diamond-cubic crystal structure
- Bandgap approximately: $E_g \approx 0.66\ eV$
- at room temperature.
- Germanium has relatively high carrier mobility.

Advantages

- High carrier mobility
- Useful for high-speed applications
- Useful in infrared technology

Disadvantages

- Smaller bandgap
- Higher leakage current
- Less suitable for many high-temperature applications than silicon

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- **Gallium Arsenide:** Gallium arsenide is a compound semiconductor consisting of:
- $Ga + As$
- It is commonly abbreviated as: $GaAs$.
- Its bandgap is approximately: $E_g \approx 1.42\ eV$
- at room temperature.
- GaAs has high electron mobility and is particularly useful for high-frequency and optoelectronic applications.

Applications

- Microwave circuits
- Satellite communication
- Radar
- High-frequency amplifiers
- LEDs
- Laser diodes
- Photodetectors
- Solar cells

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Comparison of Si, Ge and GaAs

Property	Silicon	Germanium	GaAs
Symbol	Si	Ge	GaAs
Type	Elemental	Elemental	Compound
Valence electrons	4	4	III-V compound
Bandgap	~1.12 eV	~0.66 eV	~1.42 eV
Crystal structure	Diamond cubic	Diamond cubic	Zinc blende
Electron mobility	Moderate	High	Very high
Thermal stability	Good	Lower	Good
Cost	Relatively low	Higher	High
IC manufacturing	Excellent	Limited	Specialized
High-frequency use	Good	Good	Excellent
Optical applications	Limited	Good	Excellent

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Energy Band Concept: The electrical behavior of a semiconductor can be explained using **energy bands**. When many atoms come together to form a solid, the discrete energy levels of individual atoms spread into closely spaced energy states called **energy bands**.

The two most important bands are:

- **Valence Band:** The valence band contains electrons involved in bonding.
- **Conduction Band:** The conduction band contains electrons that can move through the material and contribute significantly to electrical conduction.

Between them is the:

Forbidden *energy gap* or Bandgap E_g

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Energy Bandgap: The bandgap is the energy difference between the conduction band and valence band.

$$E_g = E_C - E_V$$

where:

E_C =conduction-band energy

E_V =valence-band energy

The bandgap is normally measured in electron-volts (eV).

Why Bandgap Is Important: The bandgap strongly affects semiconductor behavior. A smaller bandgap generally makes it easier for electrons to gain sufficient energy to enter the conduction band. A larger bandgap makes it more difficult.

This influences:

- ❑Electrical conductivity
- ❑Temperature dependence
- ❑Optical absorption
- ❑Light emission
- ❑Leakage current
- ❑Breakdown behavior
- ❑Device operating temperature

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- **Direct and Indirect Bandgap Semiconductors:** Semiconductors can be classified according to the relationship between the valence-band maximum and conduction-band minimum.
- **Direct Bandgap:** In a direct bandgap semiconductor, electrons can transition between the conduction and valence bands without a significant change in momentum. Examples: GaAs, GaN, InP

These materials are suitable for light-emitting devices.

✓Applications include:

✓LEDs

✓Laser diodes

✓Optical communication

Indirect Bandgap: In an indirect bandgap semiconductor, a momentum change is required during the transition. Examples:

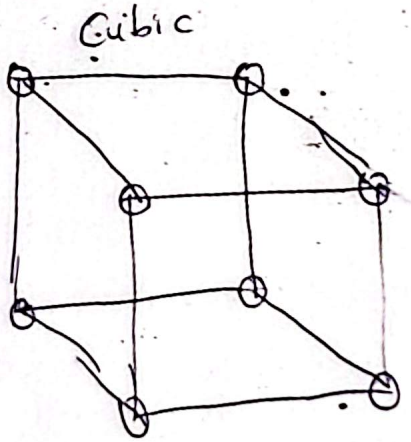
✓Silicon

✓Germanium

Silicon is therefore not an efficient light emitter.

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- **Semiconductor Crystal Structure** A semiconductor is not simply a collection of independent atoms. Atoms are arranged in an ordered three-dimensional pattern called a **crystal lattice**.
- A crystal structure determines many physical properties of a semiconductor.
- SImportant concepts include:
  - ✓Lattice
  - ✓Unit cell
  - ✓Crystal planes
  - ✓Crystal directions
  - ✓Miller indices
  - ✓Atomic bonding



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Crystal Lattice

- A **crystal lattice** is a regular, repeating arrangement of atoms or points in space.
- A semiconductor crystal can be described by repeating a basic structural unit known as the **unit cell**.
- The unit cell contains sufficient information to reproduce the entire crystal by translation.

Silicon Crystal Structure

- Silicon has a **diamond cubic crystal structure**.
- The diamond structure can be regarded as two interpenetrating face-centered cubic structures displaced relative to each other.
- Each silicon atom has four nearest-neighbor silicon atoms.
- The bonding is predominantly covalent.
- This structure is responsible for many of silicon's useful semiconductor properties.

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Germanium Crystal Structure

- Germanium also has the diamond cubic crystal structure.
- Each germanium atom forms four covalent bonds with neighboring germanium atoms.
- Therefore, silicon and germanium have similar basic bonding arrangements.
- However, their atomic sizes and electronic properties differ.

GaAs Crystal Structure

- GaAs generally forms the **zinc-blende crystal structure** under ordinary conditions.
- It consists of two interpenetrating face-centered cubic sublattices, one associated with gallium and the other with arsenic.
- The structure contributes to its important electronic and optical properties.

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- **Lattice Constant:** The **lattice constant** is a characteristic dimension of a crystal unit cell. For a cubic crystal, the lattice constant is commonly represented by:*a*and is usually measured in: metres (m) nanometres (nm) angstroms (Å)

The lattice constant is important in determining:

- ❑Atomic spacing
- ❑Crystal density
- ❑Crystal matching
- ❑Epitaxial growth
- ❑Semiconductor heterostructures

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- Miller indices provide a standard notation for crystallographic planes and directions.
- Plane: (h k l).
- Family of equivalent planes: {h k l}.
- Direction: [u v w].
- Family of equivalent directions: <u v w>

They are important in wafer orientation, crystal growth, etching, diffraction and device fabrication.

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Miller indices

Procedure:

- ☐Determine the plane's intercepts on x, y and z axes in lattice units.
- ☐Take the reciprocals of the intercepts.
- ☐Multiply by a common factor to obtain the smallest integers.
- ☐Write the result as (h k l).
- ☐An intercept of ∞ gives an index of 0 because the plane is parallel to that axis.
- ☐Negative indices are shown with a bar, e.g. (1̄10).

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EXAMPLE

- Plane intercepts: a, 2b and 3c.
  - Intercepts in lattice units = (1, 2, 3).
  - Reciprocals = (1, 1/2, 1/3).
  - Multiply by 6 = (6, 3, 2).
  - Therefore Miller indices = (632).
  - Remember: Miller indices are obtained from reciprocals of intercepts, not the intercepts themselves.
- Plane intercepts x = a, y = 2b and z = ∞.
  - Intercepts = (1, 2, ∞).
  - Reciprocals = (1, 1/2, 0).
  - Multiply by 2 = (2, 1, 0).
  - Therefore the Miller indices are (210).
  - The zero indicates that the plane is parallel to the z-axis.

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Revision Question

- ❑ Define a semiconductor.
- ❑ Distinguish between a conductor, semiconductor and insulator.
- ❑ What are valence electrons?
- ❑ Why does silicon have four valence electrons?
- ❑ Explain covalent bonding in silicon.
- ❑ Define intrinsic semiconductor.
- ❑ Define extrinsic semiconductor.
- ❑ What is doping?
- ❑ Differentiate between N-type and P-type semiconductors.
- ❑ What is a donor impurity?

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- What is an acceptor impurity?
- Define energy bandgap.
- Differentiate between direct and indirect bandgap semiconductors.
- State the crystal structure of silicon.
- State the crystal structure of GaAs.
- What are Miller indices?
- Why is silicon widely used in integrated circuits?
- State three applications of GaAs.
- Define carrier mobility.
- Explain why semiconductor conductivity increases with temperature.

Calculation Questions

- An n-type semiconductor has an electron concentration of $5 \times 10^{16} cm^{-3}$ .If the electron mobility is $1200 cm^2/Vs$,calculate its conductivity, assuming hole conduction is negligible.

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- A semiconductor has:

$$n = 2 \times 10^{16} cm^{-3}$$
$$p = 5 \times 10^{15} cm^{-3}$$
$$\mu_n = 1300 cm^2/Vs$$
$$\mu_p = 450 cm^2/Vs$$

Calculate its conductivity using: $\sigma = q(n\mu_n + p\mu_p)$

- Explain why GaAs is preferred to silicon for some high-frequency applications.
- A crystal plane has intercepts $2a$, $3b$, and ∞c. Determine its Miller indices.
- Explain the importance of semiconductor bandgap in electronic and optoelectronic device design.

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